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Shuxian Jin, Vanessa Clemens, Laura C. Hoenig, Giuliana Spadaro, The Cross-Cultural Cooperation Network.

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Advancing Global Social Science through Cross-Societal Online Behavioral Economic Games (CoBEx): Lessons Learned and A Practical Guide

Shuxian Jin^{1,2}, Vanessa Clemens³, Laura C. Hoenig⁴, Carsten K.W. De Dreu⁵, Angela Dorrough^{6,3}, Catherine Molho⁷, Marina Orifici^{6,3}, Angelo Romano⁸, Shaul Shalvi⁹, Ayse K. Uskul^{2,10}, Giuliana Spadaro⁴

¹ School of Psychological and Cognitive Sciences, Key Laboratory of Machine Perception (Ministry of Education), and Beijing Key Laboratory of Brain-Computer Interface and Mental Health Modulation, Peking University

² School of Psychology, University of Sussex

³ The Social Cognition Center Cologne, University of Cologne

⁴ Department of Experimental and Applied Psychology, Vrije Universiteit Amsterdam

⁵ Faculty of Behavioural and Social Sciences, University of Groningen

⁶ Department of Psychology, Behavioral Economics and Intercultural Psychology, FernUniversität in Hagen

⁷ Department of Social and Behavioral Sciences, Toulouse School of Economics, University of Toulouse Capitole

⁸ Social, Economic and Organisational Psychology Department, Leiden University

⁹ Amsterdam School of Economics, University of Amsterdam

¹⁰ Faculty of Arts and Social Sciences, Sabanci University

Corresponding Authors: Dr. Shuxian Jin, School of Psychological and Cognitive Sciences, Key Laboratory of Machine Perception (Ministry of Education), and Beijing Key Laboratory of Brain-Computer Interface and Mental Health Modulation, Peking University, No. 5, Summer Palace Road, Haidian District, Beijing 100871, China, Email: s.jin@pku.edu.cn

Abstract

Understanding human behavior across diverse cultural settings is essential for advancing theories that address pressing global social challenges. Recent advances, which use an innovative methodology—*cross-societal online behavioral economic games* (CoBEx)—have revealed key insights into cross-cultural differences and similarities in the mechanisms and functions of human behavior. To foster this emerging field, we discuss the key successes and lessons learned from ten years of experience in designing and conducting CoBEx studies. We illustrate the theoretical and methodological benefits of CoBEx, and offer practical recommendations on key design considerations, study materials preparation, ethics and funding, as well as analysis, interpretation, and dissemination processes. This paper aims to equip researchers around the globe with the necessary tools to conduct methodologically rigorous and culturally informed CoBEx studies, ultimately advancing the field of psychological science and beyond.

Keywords: cross-cultural research, behavioral experiments, online methods, economic games, global science

Understanding human behavior across societies and diverse cultural settings is crucial for advancing social science theories and for designing tailored policy interventions in an increasingly interconnected world. Examining how individuals from various cultural and societal backgrounds think, feel, and behave can uncover patterns that may be otherwise neglected within homogeneous populations (Krys et al., 2025). While the importance of conducting research across different cultural contexts has been widely recognized in the social sciences, running behavioral experiments in laboratory settings across societies presents significant logistical and practical challenges. These include accessing comparable samples from the general population, training local research assistants, achieving methodological comparability across sites, and arranging translations, local ethics applications and travel, among others.

In the last decade, several research teams have started examining social behaviors in highly controlled online settings across a wide range of societies. This approach, which we term *cross-societal online behavioral economic games* (CoBEx), addresses many of the above-mentioned challenges and offers unique theoretical and methodological advantages. Recent applications of CoBEx have provided valuable insights into cross-cultural similarities and differences in the mechanisms and functions of human behavior (Table 1 and Figure 1). Here, we share the key successes and lessons learned from these ten years of experience leading and collaborating on CoBEx studies, and illustrate their potential to advance global behavioral and psychological science.

Table 1. A non-exhaustive list of recent published research using CoBEx

#	CoBEx study	Behavioral paradigm	Outcome	Interaction	Incentivized	<i>k</i> of societies	List of societies*	<i>N</i> of participants	Link to pre-registration/ data/materials
1	(Dorrough & Glöckner, 2016)	Prisoner's dilemma game	Cooperation; Expectation of others' cooperation	cross-society	All societies	Study 1: 6; Study 2: 10	Study 1: Germany, India, Israel, Japan, Mexico, United States; Study 2: Afghanistan, Bangladesh, France, Germany, India, Israel, Japan, Mexico, Spain, United States	Study 1: 1,227 (51% female; 18-83 years of age)†; Study 2: 485 (33% female; 18-61 years of age)	https://osf.io/phgbs
2-1	(Romano et al., 2017)	Trust game	Trust behavior; Trustworthiness; Expectations in the trust game	cross-society; within-society	Only in South Korea and United Kingdom	17	Argentina, Brazil, China, Estonia, Germany, Indonesia, Italy, Japan, New Zealand, Poland, Russia, South Korea, Spain, Taiwan, Turkey, United Kingdom, United States	3,236 [47% female; M_{age} (SD) = 44.12 (13.53)]	https://osf.io/r8kwt/
2-2	(Spadaro et al., 2024)								https://osf.io/acsg2/
3	(Fiedler et al., 2018)	Dictator game	Prosocial behavior	cross-society; within-society	All societies	5	Latin American countries (Chile, Peru, Colombia, Venezuela), United States	915 (49% female; 18–84 years of age, M_{age} = 40.47)	https://osf.io/76bzbq/
4	(Dorrough & Glöckner, 2019)	Prisoner's dilemma game	Cooperation; Expectation of others' cooperation	cross-society; within-society	All societies	12	Subsample 1: Australia, Austria, Chile, England, Russia, Venezuela; Subsample 2: Germany, India, Israel, Japan, Mexico, United States	Subsample 1: 1,202 (49% female; 18-83 years of age); Subsample 2: 1,227 (51% female; 18-83 years of age)†	https://osf.io/ruj39/
5-1	(Romano, Sutter, Liu, Yamagishi, et al., 2021)	Prisoner's dilemma game	Cooperation; Expectation of others' cooperation	cross-society; within-society	Manipulated in Brazil, India, and Poland	42	Argentina, Australia, Bolivia, Brazil, Canada, China, Colombia, Egypt, Finland, Germany, Greece, Hong Kong, Hungary, India, Indonesia, Italy, Japan, Kenya, Mexico, Malaysia, Morocco, Netherlands, New Zealand, Nigeria, Pakistan, Panama, Peru, Philippines, Poland, Portugal, Russia, Serbia, Singapore, South Africa, South Korea, Spain, Sweden, Taiwan, Turkey, Venezuela, United Kingdom, United States	18,411 [53% female; M_{age} (SD) = 37.40 (12.52)]	https://osf.io/68wds/
5-2	(Romano, Sutter, Liu, & Balliet, 2021)								
7-1	(Dorrough & Glöckner, 2021)	Prisoner's dilemma game	Cooperation; Expectation of others' cooperation	within-society	All societies	10	Chile, China, Colombia, Indonesia, Japan, Mexico, Russia, Spain, Sweden, United States	1,903	https://osf.io/xjh7z/

7-2	(Dorrough et al., 2021) (Study 1)	Compensation-punishment game	Prosocial punishment; Prosocial compensation				1,791	https://osf.io/re7n3/
7-3	(Olsson et al., 2021)	Prisoner's dilemma game; Compensation-punishment game	Cooperation; Expectation of others' cooperation; Prosocial punishment; Prosocial compensation				1,915 (52% female; 18-87 years of age)	https://osf.io/qbp87/
8-1	(Romano et al., 2022)	Attacker-defender game	Competition; Defense	cross-society; within-society	All societies	51	12,863 [50% female; $M_{age}(SD) = 38.86 (12.29)$]	https://osf.io/kuc45/
8-2	(Romano et al., 2024) (Study 1)							
9	(Dorrough et al., 2023)	Bribery game	Bribe offer and acceptance decisions; Expectation of others' bribe offer and acceptance decisions	cross-society; within-society	All societies	Pilot: 3; Main: 18	Pilot: 890; Main: 5,582	https://osf.io/xc9qf/
10	(Molho et al., 2025)	Dictator game ('willful ignorance' paradigm)	Prosocial choices	within-society	All societies	20	7,978 [49.1% female; $M_{age}(SD) = 39.3 (12.9)$]	https://osf.io/pn2kx/
12	(Clemens et al., 2025)	Social value orientation slider measure	Prosociality	cross-society	All societies	Study 1: 1; Study 2: 25	Study 1: 336 [44% female; $M_{age}(SD) = 48.38 (16.5)$]; Study 2:	https://osf.io/s43na/

							India, Indonesia, Japan, Kenya, Morocco, New Zealand, Norway, Romania, Saudi Arabia, Singapore, Slovakia, South Africa, Sweden, Switzerland, Thailand, Turkey, United Arab Emirates, United States	6,182 [50% female; M_{age} (SD) = 42.64 (15.4)]	
11	(Jin et al., 2026)	Contest game; Step-level public goods game	Competition; Expectation of other's competition; Cooperation; Expectation of other's cooperation; Coordination decisions	within-society	All societies	13	Cyprus (both Greek and Turkish Cypriot communities), Egypt, Greece, Italy, Lebanon, Morocco, Spain, Turkey, Japan, South Korea, United Kingdom, United States	3,371 [50% female; M_{age} (SD) = 40.79 (14.36)]	https://osf.io/3dscw/

Note. Interaction = whether the interaction took place between participants from the same society (within-society interaction) or different societies (cross-society interaction); Incentivized = whether participants received actual incentives for their decisions in the game. *We use the term “society” to inclusively refer to the countries, nations, and/or special administrative regions (e.g., Hong Kong) investigated in previous CoBEx studies. †Study 1 of Dorrough & Glöckner (2016) and subsample 2 of Dorrough & Glöckner (2019) were based on the same sample.

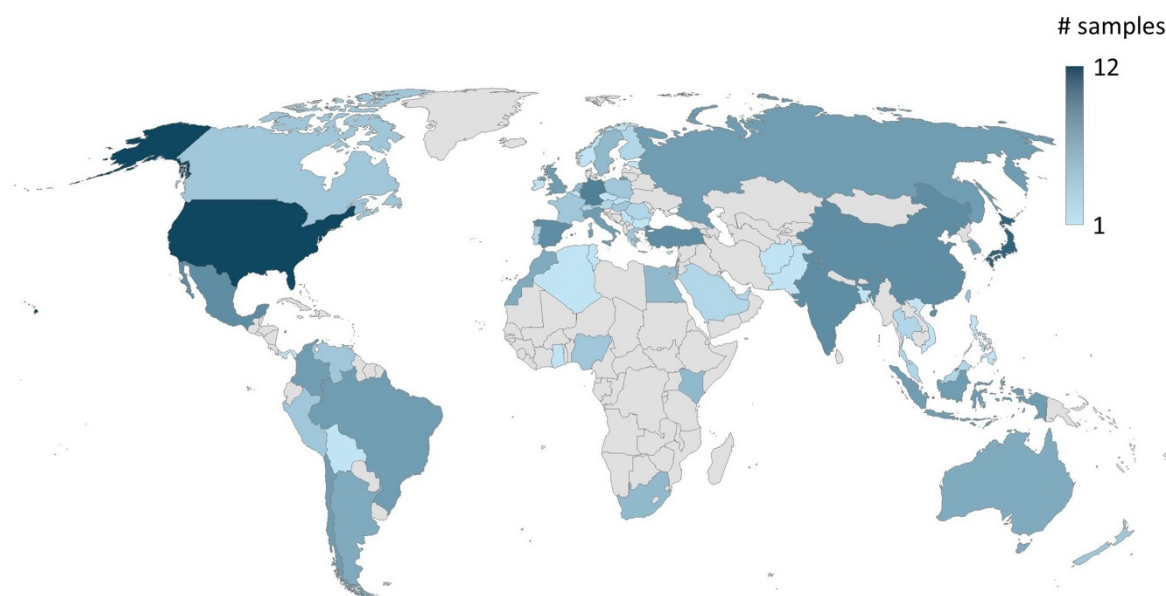


Figure 1. Number of samples (1-12) for each of the 66 societies covered by existing CoBEx studies included in Table 1.

Key advantages of CoBEx

Theoretical advantages

Social science theories aim to explain both universalities and variation in social behavior across contexts. Relevant phenomena may relate to broadly shared global challenges (e.g., cooperation, corruption) or to more context-specific challenges. Yet, the patterns these theories propose have largely been tested in a small subset of societies and cultural contexts, predominantly sampled from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations (Apicella et al., 2020). The CoBEx approach offers key benefits for testing, refining, and extending social science theories towards a more global science. First, CoBEx studies allow researchers to test whether established theoretical propositions *generalize across diverse societies*. For example, social identity theory and bounded generalized reciprocity predict greater cooperation with ingroup than outgroup members (i.e., ingroup favoritism; Balliet et al., 2014). Leveraging national ingroups and outgroups, Romano et al. (2021) demonstrated the pervasiveness of ingroup favoritism across 42 societies. Beyond such behavioral regularities, theories also propose cross-societal variation

in the psychological mechanisms sustaining prosocial behavior (e.g., guilt vs. shame). Addressing this question, Molho et al. (2025) found consistent evidence for guilt-driven, but not shame-driven, prosociality across 20 societies. Second, CoBEx studies can help identify cultural factors that *explain meaningful behavioral variation*, thereby enriching existing theories. For instance, drawing on existing CoBEx and large-scale survey data, Eriksson et al. (2021) found that societies emphasizing tolerance tend to show stronger cooperation and cooperative norms. More recently, Jin et al. (2025) examined personally endorsed and perceived normative values of honor across 13 societies, showing how the cultural logic of honor is associated with competition and cooperation at both societal and individual levels.

Methodological advantages

The key value of CoBEx lies in the combination of incentivized economic games, online recruitment, and cross-societal sampling, which together create unique opportunities for cross-cultural research. Incentivized economic games provide a controlled way to assess consequential social behavior (Thielmann et al., 2021). Because the basic rules of the game can be held constant across contexts, systematic differences and commonalities in behavior can inform the examination of variation in beliefs, preferences, or norms across societies. Economic games are also relatively less language-dependent than many self-report measures and allow incentives to be adapted to local income levels. Online recruitment extends these advantages by overcoming the geographical, physical, and socio-economic constraints of laboratory experiments, and enabling access to larger, more diverse participant pools beyond student samples that characterized previous cross-cultural studies (Herrmann et al., 2008). In combination with cross-societal sampling, this increases the generalizability of findings across societies and captures social and cultural variation within societies (Deffner et al., 2022). CoBEx also allow researchers to study (typically asynchronous) interactions within

and across cultures (Dorrough et al., 2022), enriching the study of intercultural and intergroup behavioral dynamics.

Recommendations for designing CoBEx

CoBEx bring new opportunities to study social decision-making around the globe, while also raising challenges that warrant further discussion. Producing valid findings in cross-cultural research necessitates careful attention to design, materials, translations, ethics, and interpretation. To support this emerging field, we outline four key aspects of methodological considerations in designing and conducting CoBEx (Table 2). While not exhaustive of all circumstances, our recommendations draw on the experiences of multiple research groups who employ this approach to address fundamental questions on trust, cooperation, conflict, and corruption.

Table 2. A summary of recommendations for designing CoBEx

Study design considerations
• Select societies based on specific research objectives (e.g., generalizability, testing cultural effects) • Acquire sample through online panel providers and/or by collaborating with local researchers • Plan sample size at both society and participant levels, aligned with inferential and analytic goals • Conduct a priori or sensitivity power analyses to determine the sample size within each society • Use (a combination of) post hoc matching of decisions and/or the strategy method to enable behavioral incentivization • Ensure equivalence and consistency in incentivizing behaviors across societies
Study materials preparation
• Combine textual and visual elements in presenting game instructions • Implement comprehension check questions with feedback to improve game understanding • Adapt general quality check indicators to CoBEx studies (e.g., attention check, response time, bot detection) • Select the most feasible translation approach (e.g., forward-back translation, committee, team, AI-assisted)
Ethics & funding
• Adhere to the strictest research ethics (incl. data protection regulations) among the data collection societies • Improve equitable collaboration (e.g., open and flexible collaborative agenda, fair acknowledgement of contributions) • Consider collaborative funding opportunities (e.g., PI-led grants, consortium grants, individual resources)
Analysis, interpretation & dissemination
• Choose analytic strategies based on inferential goals and conduct robustness checks (see Table S2) • Balance the emphasis on differences and similarities across societies • Be cautious in interpreting correlational findings as evidence of causality • Acknowledge limitations (if any) when discussing the generalizability of results • Broadcast findings beyond academic audiences and English-speaking communities

Study design considerations

Society selection. We outline two approaches for selecting societies depending on research objectives. First, when the goal is to test the generalizability of psychological phenomena, we recommend sampling as many societies as possible from diverse world regions and across a broad range of societal-level dimensions (e.g., GDP, cultural values)¹. Second, when the goal is to examine phenomena theoretically linked to specific societal factors, we recommend selecting societies to maximize variability in the relevant indicator. For example, studies examining whether sex differences in cooperation vary with society-level gender inequality may benefit from sampling societies that differ substantially on indicators such as the Gender Inequality Index (Dorrough & Glöckner, 2021) (See Table S1 in the Supplementary Materials for additional society selection rationale and examples).

Sample acquisition. Participants recruitment strategies include using online panel providers, collaborating with local researchers, or a combination of both. Online panels enable access to diverse participant pools and allow sample stratification by geographic (e.g., region of residence) or demographic factors (e.g., age, gender) using quotas. We recommend that researchers establish specific prescreening and inclusion criteria (e.g., proportion of lifetime spent in the respective society) to enhance sample comparability across the societies included in their study. Collaborations with local academic institutions, though coordination-intensive, provide access to student samples that have been argued to be more socio-economically comparable across societies (Herrmann et al., 2008).

Sample size determination. CoBEx data structure typically involve decisions nested within participants and societies, making multilevel modeling an appropriate analytical approach. In such models, sample size planning should consider both the number of societies and the number of participants within each society. Statistical power for society-level effects is primarily determined by the number of societies (the highest-level cluster), whereas power

for individual-level effects depends more on the number of participants within societies. There is no universal minimum number of societies, as the required number can depend on the inferential goal, expected effect size, between-society heterogeneity, and planned analytic model. A small number of societies may be informative for descriptive analyses or fixed-effect comparisons of the included societies, but larger number (20 or more) are generally preferable when estimating society-level predictors, moderators, random effects, or cross-level interactions (McNeish & Stapleton, 2016). Researchers are encouraged to conduct a priori power analyses and/or sensitivity analyses based on their specific hypotheses and feasible sample sizes (Arend & Schäfer, 2019).

Behavior incentivization. Economic games allow attaching real consequences to choices—through monetary or non-monetary payments—and make it costly to engage in behaviors misaligned with participants’ true preferences (Thielmann et al., 2021). This advantage has motivated most CoBEx studies to incentivize participants’ decisions (and sometimes also beliefs) through additional decision-dependent bonuses², which also introduces implementation challenges in calculating said bonuses and ensuring incentive equivalence across societies. Because economic games often involve interdependent outcomes among players, researchers need to match participants’ decisions to determine their bonuses. We recommend three approaches: (a) post hoc matching, where decisions are matched after data collection; (b) the strategy method, where participants make decisions for all possible scenarios and one is randomly selected for bonuses calculation; and (c) a combination of both, such as collecting decisions in a pre-study to be asynchronously matched with decisions in the CoBEx study.

To maximize incentive equivalence, we recommend setting a conversion rate between the experimental (e.g., game tokens) and local currencies using metrics that reflect average income levels or purchasing power. For example, one token could equal a specific number of

minutes of average wage, or the cost of a common good such as a Big Mac meal (based on the Big Mac Index, <https://www.economist.com/big-mac-index>). Incentives do not always need to be monetary or provided for every decision or participant; alternatives include gift cards, vouchers, or lotteries. However, whether different types of non-monetary incentives and lottery designs are equally valued or appropriate across cultures remains underexplored.

Study materials preparation

Game instructions. Clear and extensive instructions are essential to ensure participants fully comprehend the game. While text-based instructions are the most frequently used materials for explaining rules of the game, we recommend supplementing them with infographics to illustrate the dynamics and strengthen any manipulations (e.g., the nationality of the interaction partner; see Figure 2 for an example). The combination of textual and visual elements can help improve participants' engagement and understanding of the game, reduce ambiguity arising from linguistic, translational, or cultural differences, and thereby promote more comparable comprehension across societies.³

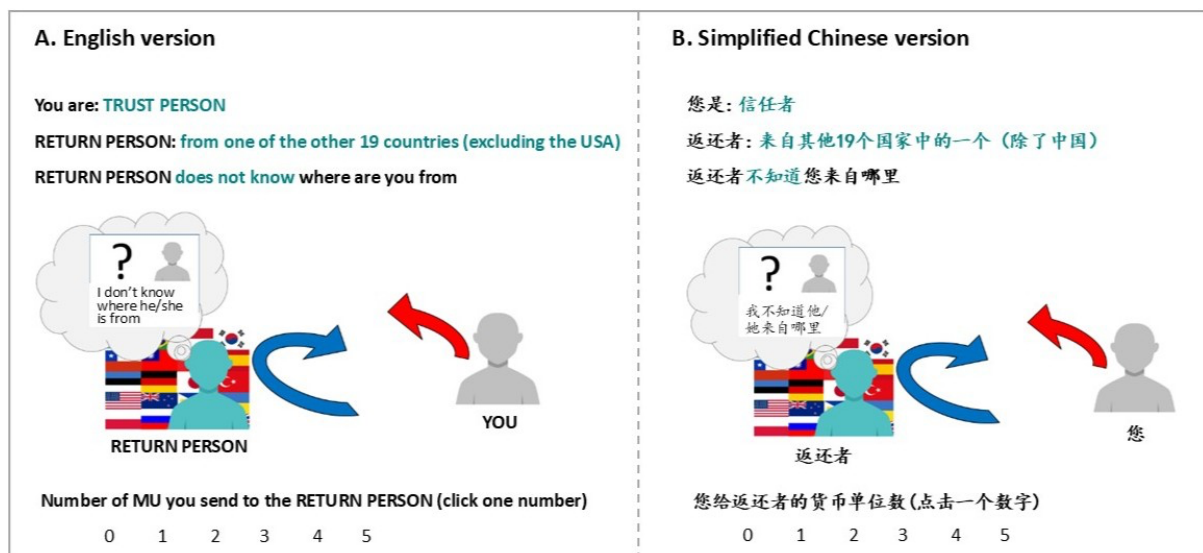


Figure 2. Example game instructions in English and Simplified Chinese illustrating a manipulation of the interaction partner's nationality (Romano et al., 2017).

Note. Full materials and translations from Romano et al. (2017) are publicly available at <https://osf.io/r8kwt/>.

Comprehension checks. Comprehension check questions can be developed to assess participants' understanding of the game rules⁴. Although some behavioral experiments in laboratory settings use open-ended questions that require mathematical calculations, we recommend simplifying such questions whenever possible. For example, multiple-choice questions with live feedback—and allowing a second attempt if answers are incorrect—can more effectively record and improve participants' understanding before they proceed to make actual decisions.

Translations. Conventional translation approaches in cross-cultural research include forward-back translation, often supported by domain experts or translation agencies. For CoBEx studies, involving domain experts can be particularly helpful because economic games contain nuances that require careful handling across languages. More recently, AI-assisted translation (e.g., ChatGPT, Claude), when combined with human quality checks, has been shown to achieve quality comparable to human translation as rated by domain experts (Kunst & Bierwiazzonek, 2023). Building on these developments, domain experts in cultural psychology have also developed tools to automate forward-back translation processes (e.g., the R Shiny APP *LLMTranslate*; see <https://cran.r-project.org/package=LLMTranslate>). Regardless of the approach chosen, we recommend transparently recording and reporting the translation process including the tools and steps involved⁵.

Ethics and funding

Ethics. Ethical considerations in CoBEx studies concern participants, collaboration, and reporting. Because data protection policies vary across societies, researchers should adhere to the strictest applicable standard to ensure full compliance (see “Behaviour incentivization” for considerations related to participant compensation). To improve equitable

collaboration and balanced power relations, researchers are encouraged to engage collaborators from diverse cultural backgrounds and adopt open, participatory research practices, such as co-determining study designs, incorporating local perspectives and resources, and pursuing subprojects within the larger collaboration (Urassa et al., 2021). Fair recognition of collaborators' contributions should also be ensured through authorship credit, data access, financial compensation, and/or acknowledgment. Finally, CoBEx findings should be reported responsibly and interpreted with caution, given the risk of misrepresentation or unintended harm (see “Interpretation and dissemination”).

Funding opportunities. Conducting CoBEx studies often requires researchers to secure substantial funding, with budgets varying based on the number of societies and participants involved, incentives, and the sample acquisition approach (e.g., panel providers can be costly). Researchers can apply for individual or consortium grants that support large-scale, collaborative projects (e.g., Psychological Science Accelerator). In grant proposals, we recommend emphasizing the unique value of the CoBEx approach in addressing specific research questions. However, funding opportunities are often limited and unequally distributed across world regions (Urassa et al., 2021). For early-career researchers or scholars based in low- and middle-income societies, a bottom-up approach—pooling individual funding sources from local collaborators across institutions—can foster larger collaborative research initiatives.

Analysis, interpretation, and dissemination

Analysis. CoBEx studies involve several analytic challenges that require careful consideration. These can include identifying meaningful between-society variation, choosing between fixed-effects and mixed-effects approaches depending on inferential goals, interpreting society-level predictors and cross-level interactions when the number of societies is limited, assessing sensitivity to influential societies, alternative model specifications, and

alternative society-level indicators, managing degrees of freedom when testing many cultural, economic, or institutional indicators; distinguishing confirmatory from exploratory analyses. We expand on these challenges and suggest analytic strategies in Table S2 in the Supplementary Materials.

Interpretation and dissemination. When interpreting and communicating behavioral findings, researchers should balance the emphasis on differences and similarities across societies. Overemphasizing differences may inadvertently reinforce cultural stereotypes, especially when it comes to socially (un)desirable behaviors such as conflict, aggression, or corruption. Visual tools can effectively illustrate between-societal (e.g., world maps, Cleveland dot plots) and within-society variation (e.g., ridgeline plots, raincloud plots), but researchers should be cautious, as simply ranking societies on behaviors—such as levels of corruption—may exaggerate or misrepresent cross-societal patterns. Similarly, when interpreting and disseminating findings from correlations between societal- or individual-level indicators and behavior, we recommend clearly distinguishing between results that can support causal interpretations, such as experimentally manipulated effects, and results that should be interpreted as correlational, such as associations involving measured societal- and individual-level predictors. Researchers should also acknowledge limitations related to the generalizability of CoBEx findings as online samples are limited to individuals with internet access and are therefore not fully representative within each society⁶. For broader impact, researchers may provide summaries of findings on project websites⁷, translate them into the primary languages of the societies studied and collaborate with local media to prepare culturally adapted press releases.

Concluding remarks and future directions

CoBEx offer a methodological innovation that enables social scientists to test behavioral theories and examine the psychological and cultural mechanisms underlying

human behavior at a global scale. By enabling tests of both universality and variability, as well as the identification of novel phenomena, CoBEx provide a powerful tool for advancing social science theories. Drawing on our teams' collective experience, we outlined key considerations and practical recommendations for designing CoBEx studies. Moving forward, researchers can further harness its potential by (a) developing formal (rather than purely verbal) theoretical specifications to test in future CoBEx studies, (b) linking behaviors measured in economic games to everyday-life experiences and real-world dynamics (e.g., Clemens et al., 2025), and (c) applying machine-learning techniques trained on existing CoBEx datasets to generate predictions for societies where direct data collection is difficult. Ultimately, we encourage researchers to realize the CoBEx' full potential while building more inclusive, collaborative, and cumulative global social sciences.

Recommended Reading

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Author contributions

Shuxian Jin played a lead role in conceptualization, project administration, visualization, and writing—original draft, and an equal role in writing—review and editing. Vanessa Clemens played a lead role in conceptualization, a supporting role in writing—original draft, and an equal role in writing—review and editing. Laura C. Hoenig played a lead role in conceptualization, a supporting role in writing—original draft, and an equal role in writing—review and editing. Giuliana Spadaro played a lead role in conceptualization, a supporting role in project administration and writing—original draft, and an equal role in writing—review and editing.

The Cross-Cultural Cooperation Network:

Carsten K.W. De Dreu	Faculty of Behavioural and Social Sciences, University of Groningen, Groningen, The Netherlands	Conceptualization (supporting); Writing – Review & Editing (equal)
Angela Dorrough	Department of Psychology, Behavioral Economics and Intercultural Psychology, FernUniversität in	Conceptualization (lead); Funding Acquisition (equal); Writing – Original Draft

	Hagen, Hagen, Germany; Department of Psychology, University of Cologne, Cologne, Germany	(supporting); Writing – Review & Editing (equal)
Catherine Molho	Department of Social and Behavioral Sciences, Toulouse School of Economics, University of Toulouse Capitole, Toulouse, France	Conceptualization (lead); Funding Acquisition (equal); Writing – Original Draft (supporting); Writing – Review & Editing (equal)
Marina Orifici	Department of Psychology, Behavioral Economics and Intercultural Psychology, FernUniversität in Hagen, Hagen, Germany; Department of Psychology, University of Cologne, Cologne, Germany	Conceptualization (supporting); Writing – Review & Editing (equal)
Angelo Romano	Social, Economic and Organisational Psychology Department, Leiden University, Leiden, The Netherlands	Conceptualization (lead); Writing – Original Draft (supporting); Writing – Review & Editing (equal)
Shaul Shalvi	Amsterdam School of Economics, University of Amsterdam, Amsterdam, the Netherlands	Conceptualization (supporting); Funding Acquisition (equal); Writing – Review & Editing (equal)
Ayşe K. Uskul	School of Psychology, University of Sussex, Brighton, The United Kingdom; Faculty of Arts and Social Sciences, Sabanci University, Istanbul, Turkey	Funding Acquisition (equal); Writing – Review & Editing (equal)

Note

¹ In recent CoBEx studies, participants' society of residence has most commonly served as the cultural unit of analysis. Accordingly, we use "society" as an example throughout this paper. However, other forms of cultural units can also be employed, referring to groups or communities sharing distinct beliefs, values, norms, and practices, which can serve as a basis for comparison or analysis (Matsumoto & Van De Vijver, 2010). Such units may be defined by socio-ecological factors (e.g., geographical location, economic subsistence) or by background/identity-related factors (e.g., religious or linguistic group membership).

² In psychology, deception is sometimes used to increase experimental control, and vignette-based designs are useful for examining individuals' judgments, expectations, and intended responses. However, these methods typically do not attach real consequences to participants' choices, which limits the extent to which observed responses can be interpreted as behavior. Incentivized economic games offer important advantages for studying consequential and strategic choices (Thielmann et al., 2021). Despite the greater logistical complexity involved, we encourage researchers to incorporate behavioral incentivization into CoBEx studies whenever feasible.

³ Standardized game instructions, experimental implementation, and translation procedures can improve procedural comparability, but they may not guarantee that the same behavioral choice has the same psychological meaning across societies. For instance, the same cooperative decision across societies may reflect different combinations of trust, norm compliance, strategic uncertainty, risk preferences, or experimental engagement. Such variation can be theoretically informative rather than only a threat to comparability. Researchers can examine this issue by measuring theoretically relevant psychological constructs, testing whether these constructs show comparable or society-specific associations with behavioral choices, and assessing participants' motivation or interpretation of the behavior through open-ended questions.

⁴ In addition, several general quality check strategies can also be adapted to CoBEx studies for assessing data quality: for example, embedding multiple attention check questions at both the decision-making and questionnaire stages, and tracking the time participants spent reading game instructions to identify unusually fast movers. An emerging challenge for online studies is the possibility that participants may use AI tools to help interpret game instructions, answer comprehension questions, or generate decision recommendations. Although there is currently no definitive solution to this problem, researchers may combine multiple indicators of data quality with bot-detection features offered by panel providers or online survey platforms (e.g., Qualtrics) to reduce the risk of non-human or automated participation. Importantly, these quality-check strategies should be adapted to local contexts, as certain checks (e.g., some bogus questions or tools like Google ReCaptcha) may function differently depending on cultural or regional settings.

⁵ Relatedly, the current author team is launching an initiative to develop an open repository (for a similar example, see <https://www.theloveconsortium.org/>) where researchers can share materials and translations of economic games in multiple languages and the datasets drawn

from their own projects. These resources could enhance the replicability of behavioral experiments in cross-cultural research, provide future researchers with datasets for formulating and testing new questions, and offer tools for efficiently adapting their own CoBEx study designs.

⁶ Online samples are limited to individuals who have internet access, can be reached through online recruitment panels, are willing to participate in online studies, and pass prescreening and quality-control procedures. As a result, between-society comparisons may partly reflect differences in sampling frames, internet penetration, urbanicity, education, socioeconomic status, and familiarity with online tasks. Researchers should therefore be transparent about recruitment procedures, sample composition, exclusion criteria and rates, and the limits of generalizability within each society.

⁷ For an example, see <https://dlc-studies.uni-koeln.de/research/cross-national-social-preferences>.

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Supplementary Materials for

**Advancing Global Social Science through Cross-Societal Online Behavioral
Economic Games (CoBEx): Lessons Learned and A Practical Guide**

Table S1. Examples of society-selection strategies in previous CoBEx studies

Example CoBEx study	Research objectives	Society selection strategies
(Dorrough & Glöckner, 2021)	Examine sex differences in prosocial behavior in social dilemmas at both the individual- and societal-level, and examine potential drivers of these differences.	Selected 10 societies that vary on the Gender Inequality Index (GII) in the year of data collection 2017 (http://hdr.undp.org/en/content/gender-inequality-index-gii): Chile (GII rank 72 of 189), China (rank 36), Colombia (rank 87), Indonesia (rank 104), Japan (rank 22), Mexico (rank 76), Russia (rank 53), Spain (rank 15), Sweden (rank 3), USA (rank 41).
(Dorrough et al., 2023)	Examine whether the interaction partner's national background is associated with individual's bribery offers and acceptance in cross-societal interactions.	Selected 21 societies that vary on Transparency International's Bribe Payers Index published in 2011 (https://www.transparency.org/en/publications/bribe-payers-index-2011), including the highest-ranked and lowest-ranked societies.
(Molho et al., 2025)	Examine whether guilt- and shame-based mechanisms promote prosocial behavior across societies, and whether these effects vary as a function of individual- and society-level guilt- and shame-proneness.	Selected 20 societies that cover the full range of cross-country variation in impersonal prosociality (i.e., altruism as measured in the Global Preferences Survey, data collected in 2012, https://gps.econ.uni-bonn.de/).
(Clemens et al., 2025)	Examine variation in cross-national social preferences beyond ingroup favoritism and identify individual-, national-, and dyadic-level factors that explain prosociality toward people from different nations.	Selected 25 societies that varied on various economic, institutional, and social-ecological factors such as their cultural background as indicated by the Hofstede cultural dimensions (https://www.hofstede-insights.com/product/compare-countries/), national wealth as indicated by the GDP per capita, and institutional quality as indicated by the rule of law which were retrieved in the year of data collection 2021 from http://data.worldbank.org/indicator/ .

Table S2. Analytic challenges and suggested strategies in CoBEx studies

Analytic challenge	Why it matters in CoBEx studies	Suggested analytic strategies	Recommended reading
Identifying meaningful between-society variation	CoBEx studies often aim to examine whether behavior differs across societies. However, apparent differences may be small relative to within-society variation or may reflect noise rather than systematic between-society heterogeneity.	Report descriptive statistics and visualizations by society; estimate intraclass correlations (ICCs); use multilevel models to partition within- and between-society variance; avoid interpreting statistically significant differences without considering effect sizes and the uncertainty around the estimate.	(Lorah, 2018; Nezlek, 2010)
Choosing between fixed-effects and mixed-effects approaches	Different modeling choices serve different inferential goals. Models with society fixed effects are useful when the goal is to compare the specific societies included in a CoBEx study. By contrast, mixed-effects models estimate between-society variation by treating societies as a grouping factor and partially pooling society-level estimates. The latter approach is useful when researchers aim to characterize variation beyond the specific societies included in a CoBEx study, although such inferences depend on the number and selection of societies.	Match model choice to the research question; use fixed-effects models for society-specific comparisons; use mixed-effects models when estimating variance across societies or testing predictors at multiple levels; compare whether conclusions are robust across reasonable model specifications.	(McNeish & Kelley, 2019)
Interpreting society-level predictors with a limited number of societies	CoBEx studies may include many participants but relatively few societies. This limits statistical power for testing society-level predictors and can make estimates unstable.	Treat society-level analyses cautiously when the number of societies is small; report uncertainty clearly, for example by presenting confidence or credible intervals around society-level estimates and avoiding strong conclusions when these intervals are wide; avoid overinterpreting society-level correlations; prioritize theoretically motivated predictors; consider presenting society-level analyses as exploratory when appropriate.	(Bryan & Jenkins, 2016)

Interpreting cross-level interactions	In CoBEx studies, researchers may test whether individual-level associations vary as a function of society-level characteristics, but such interactions can be difficult to estimate reliably with few societies.	Pre-specify theoretically motivated cross-level interactions where possible; limit the number of interaction tests; visualize predicted effects; report confidence or credible intervals around estimated effects; interpret exploratory cross-level interactions cautiously.	(Aguinis et al., 2013)
Assessing sensitivity to influential societies	In CoBEx studies, a single society may sometimes strongly affect society-level associations, especially when the number of societies is limited or when one society is an outlier on a key predictor or outcome.	Conduct leave-one-society-out analyses; inspect influence diagnostics; compare results with and without potentially influential societies; report whether substantive conclusions remain stable.	(Nieuwenhuis et al., 2012)
Assessing sensitivity to alternative model specifications	In CoBEx studies, conclusions may depend on analytic choices such as covariate inclusion, distributional assumptions, treatment of exclusions, or whether society is modeled using fixed or mixed effects.	Conduct robustness checks across theoretically defensible model specifications; transparently report the primary model and alternative specifications; explain why the main model was selected.	(Simonsohn et al., 2020)
Assessing sensitivity to alternative society-level indicators	In CoBEx studies, cultural, economic, and institutional constructs can often be operationalized using multiple indicators or data sources. Different society-level indicators may produce different conclusions.	Justify the selected society-level indicators theoretically; report data sources and years; test robustness using alternative indicators where appropriate; be cautious when indicators are highly correlated or conceptually overlapping. In such cases, dimension-reduction approaches, such as principal component analysis, may be used to derive a composite indicator, provided that the theoretical meaning and limitations of the resulting component are clearly explained.	(Claessens et al., 2023)
Distinguishing confirmatory from exploratory analyses	Readers need to know whether society-level hypotheses were specified in advance or identified after inspecting the data in CoBEx studies. This is especially important in rich cross-societal datasets.	Clearly label preregistered confirmatory analyses and exploratory analyses; avoid presenting exploratory findings as definitive evidence; use exploratory results to generate hypotheses for future studies.	(Nosek et al., 2018)

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